

Week 4: Mixed Strategies

Strategic Randomisation, Expected Payoffs, and Indifference

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When predictability makes you exploitable, randomisation can be rational.

From Week 3 to Week 4

Week 3: Pure Strategies

- Mark best responses
- Find stable cells
- Allow one or several equilibria
- Recognise when no pure equilibrium exists

Week 4: Mixed Strategies

- Randomise over pure actions
- Calculate expected payoffs
- Apply indifference conditions
- Interpret equilibrium frequencies

If every predictable action can be exploited, what pattern can remain stable?

Opening Intuition: Rock, Paper, Scissors

Why No Pure Action Is Stable

Rock is beaten by Paper, Paper is beaten by Scissors, and Scissors is beaten by Rock.

- Always Rock is exploitable.
- Always Paper is exploitable.
- Always Scissors is exploitable.

Stable Behaviour

Use each action with probability $1/3$ so that the opponent cannot profit from predicting one action.

Randomisation removes the exploitable pattern.

Today's Learning Outcomes

By the end of class, you should be able to:

- 1 distinguish pure and mixed strategies;
- 2 explain why strategic randomisation can prevent exploitation;
- 3 calculate expected payoffs;
- 4 apply the indifference condition;
- 5 solve a two-by-two mixed-strategy Nash equilibrium;
- 6 construct and interpret a defensible business application.

Opening Decision: Promotion Timing

The Situation

A coffee chain chooses whether to promote on **Monday** or **Wednesday**. Its rival simultaneously chooses which day to prepare a defensive campaign.

	Rival: Prepare Mon	Rival: Prepare Wed
Promote Mon	(3, 3)	(6, 0)
Promote Wed	(5, 1)	(3, 2)

Can either player choose one day with certainty and remain stable?

First payoff: coffee chain

Second payoff: rival

Why Do These Payoffs Differ?

Business Assumptions

- Monday offers greater potential demand than Wednesday.
- A correctly prepared rival reduces the promotion's effectiveness.
- Preparing a defensive campaign is costly.
- Preparing on the wrong day wastes resources and loses customers.

Rival predicts correctly

- Mon–Mon: (3, 3)
- Wed–Wed: (3, 2)

Rival prepares on the wrong day

- Mon–Wed: (6, 0)
- Wed–Mon: (5, 1)

Monday offers the larger opportunity, but it also attracts more defensive attention.

Best Responses Cycle

	Prepare Mon	Prepare Wed
Promote Mon	(3, 3)	(6, 0)
Promote Wed	(5, 1)	(3, 2)

- 1 At Mon–Mon, the **chain** wants Wednesday.
- 2 At Wed–Mon, the **rival** wants Wednesday.
- 3 At Wed–Wed, the **chain** wants Monday.
- 4 At Mon–Wed, the **rival** wants Monday.

No cell contains both players' best responses.

Pure and Mixed Strategies

Pure Strategy

Choose one action with probability 1.

Example: always promote Monday.

Mixed Strategy

Assign probabilities to pure actions, with probabilities summing to 1.

Example: Monday with p , Wednesday with $1 - p$.

Important

Randomisation must be sufficiently unpredictable to the rival. A hidden but inferable rule can still be exploited.

Expected Payoff

Definition

Expected payoff is a probability-weighted average of the payoffs from uncertain outcomes.

$$E(u) = \text{Pr}(1)u_1 + \text{Pr}(2)u_2 + \dots$$

Simple Example

If an action gives payoff 5 with probability 0.4 and payoff 2 with probability 0.6, then

$$E(u) = 0.4(5) + 0.6(2) = 3.2.$$

The numerical payoffs must be treated as cardinal utility values for expected-payoff calculations.

The Indifference Condition

Core Principle

In a fully mixed equilibrium, every pure strategy used with positive probability gives the player the same expected payoff.

- Find the chain's probability p by equalising the **rival's** expected payoffs.
- Find the rival's probability q by equalising the **chain's** expected payoffs.
- Your mix is determined by the **other player's payoff entries**.

Why?

If one supported action gave the opponent a strictly higher expected payoff, the opponent would choose it with certainty rather than mix.

Step 1: Find the Chain's Mix p

Let p be the probability of Promote Monday.

Choose p so that the rival is indifferent between its two preparation days.

Prepare Monday

$$\begin{aligned} E(R_M) &= 3p + 1(1 - p) \\ &= 1 + 2p \end{aligned}$$

Prepare Wednesday

$$\begin{aligned} E(R_W) &= 0p + 2(1 - p) \\ &= 2 - 2p \end{aligned}$$

$$1 + 2p = 2 - 2p \quad \Rightarrow \quad p = 0.25$$

The chain uses Monday 25% and Wednesday 75%.

Step 2: Find the Rival's Mix q

Let q be the probability of Prepare Monday.

Choose q so that the **chain** is indifferent between its two promotion days.

Promote Monday

$$\begin{aligned} E(C_M) &= 3q + 6(1 - q) \\ &= 6 - 3q \end{aligned}$$

Promote Wednesday

$$\begin{aligned} E(C_W) &= 5q + 3(1 - q) \\ &= 3 + 2q \end{aligned}$$

$$6 - 3q = 3 + 2q \quad \Rightarrow \quad \boxed{q = 0.6}$$

The rival prepares Monday 60% and Wednesday 40%.

Interpret and Verify the Equilibrium

Mixed-Strategy Nash Equilibrium

- **Chain:** Promote Mon with 0.25, Promote Wed with 0.75.
- **Rival:** Prepare Mon with 0.6, Prepare Wed with 0.4.

Chain's expected payoff:

$$6 - 3(0.6) = 4.2$$

Rival's expected payoff:

$$1 + 2(0.25) = 1.5$$

Interpret Carefully

Monday is the larger opportunity, so the rival prepares Monday more often. Anticipating this heavier defence, the chain promotes more often on Wednesday.

A Reliable Mixed-Strategy Workflow

- 1 Mark pure best responses and confirm that no pure equilibrium exists.
- 2 Define p , $1 - p$, q , and $1 - q$ in words.
- 3 Use the **column player's payoffs** to solve p .
- 4 Use the **row player's payoffs** to solve q .
- 5 Check that probabilities lie between 0 and 1 and sum to 1.
- 6 Substitute back to verify indifference and calculate expected payoffs.

Common Error

Do not assume 50–50, and do not use your own payoffs to calculate your own probability.

Application 1: Design a Promotion or Positioning Game

Group Task

Create a real two-firm case in which one player wants to avoid or surprise the rival, while the rival wants to anticipate or neutralise the move.

Possible strategic dimensions:

- Promotion day or channel
- Customer segment
- Product-launch timing
- Location or offer type

Required analysis:

- Defend the payoff matrix.
- Show best-response cycling.
- Solve p and q .
- Interpret the expected payoffs.

Application 2: Inspection and Deterrence

Illustrative Structure

An inspector chooses Audit or Do Not Audit; an organisation chooses Comply or Violate.

- Auditing every case is costly.
- Never auditing may encourage violation.
- A random audit probability can create deterrence while controlling inspection cost.
- The probability depends on audit cost, penalties, detection, and the benefits of violation.

Discussion Question

Whose payoffs determine the audit probability, and whose payoffs determine the compliance probability?

Managerial Interpretation: When Is Randomisation Useful?

Potential Benefit

- Reduce competitor pre-emption
- Preserve surprise or deterrence
- Test demand across times or segments
- Prevent exploitation of a fixed pattern

Potential Cost

- Operational complexity
- Customer confusion
- Information leakage
- Legal or platform constraints

Variation alone does not prove mixing: prices may also change with demand, inventory, or new information.

Checklist for a Strong Business Application

Build the Model

- Real players and feasible actions
- Payoffs grounded in demand, cost, or market position
- No pure equilibrium established

Solve and Interpret

- Variables defined clearly
- Correct cross-player equations
- Probabilities verified by substitution
- Implementation risk identified

Take the Payoffs One Step Further

Explain where the numbers come from using customer demand, profit margins, costs, or another simple economic model.

Optional Extension: The BOGO Promotion Game

Save for the Start of Next Class

Two coffee chains choose between a Monday and Friday buy-one-get-one-free promotion. One chain wants to promote when the rival is unprepared; the rival wants to anticipate and match it.

- 1 Construct a defensible two-by-two payoff matrix.
- 2 Use a simple demand or profit calculation to justify the numbers.
- 3 Mark best responses and confirm whether mixing is required.
- 4 Define and solve both equilibrium probabilities.
- 5 Verify expected payoffs and state one implementation risk.

Week 4: Main Takeaways

- 1 A mixed strategy is a probability distribution over pure strategies.
- 2 Mixing is relevant when deterministic behaviour is unstable or exploitable.
- 3 Expected payoff is a probability-weighted average.
- 4 Supported strategies must give equal maximum expected payoffs.
- 5 Your probability is found by making the other player indifferent.
- 6 Equilibrium probabilities need not be equal or 50–50.
- 7 Business value depends on payoff foundations and implementation feasibility.

**Check pure actions. Define probabilities. Equalise the opponent.
Verify. Interpret.**

Exit Ticket and Looking Ahead

Exit Ticket

Why is your equilibrium probability calculated using the other player's payoffs? Why must the other player be indifferent?

Before Week 5

Complete the pre-reading on commitment. Come ready to ask:

How can an observable, difficult-to-reverse action change a rival's response?

Supplementary: Rock–Paper–Scissors Expected Payoff

Opponent Mixes Equally

Against probabilities $(1/3, 1/3, 1/3)$, choosing Rock gives

$$\frac{1}{3}(0) + \frac{1}{3}(-1) + \frac{1}{3}(1) = 0.$$

- Paper also gives expected payoff 0.
- Scissors also gives expected payoff 0.
- Every pure action is a best response, so the player is willing to mix.

Supplementary: Deriving the Promotion Payoffs

Illustrative Logic

Payoff = customer contribution – promotion or defence cost

Rival predicts correctly

- Mon–Mon, (3, 3): high demand, but the defence limits the promotion.
- Wed–Wed, (3, 2): the defence works, but protects a smaller opportunity.

Rival prepares on the wrong day

- Mon–Wed, (6, 0): the chain captures the larger Monday opportunity.
- Wed–Mon, (5, 1): the chain gains, but Wednesday demand is lower.

The values are illustrative; their ranking should follow from demand, margins, customer movement, and preparation costs.

Supplementary: General Two-by-Two Setup

	Column: Left	Column: Right
Row: Up	(a, e)	(b, f)
Row: Down	(c, g)	(d, h)

- Row plays Up with p ; Column plays Left with q .
- Find p from Column's indifference:

$$pe + (1 - p)g = pf + (1 - p)h.$$

- Find q from Row's indifference:

$$qa + (1 - q)b = qc + (1 - q)d.$$

Supplementary: Full Verification Checklist

- 1 Confirm that $0 \leq p, q \leq 1$.
- 2 Confirm that each player's probabilities sum to 1.
- 3 Supported strategies give equal expected payoffs.
- 4 Supported strategies achieve the maximum expected payoff.
- 5 Unsupported strategies give no higher expected payoff.

Warning

Algebra can produce a number even when the assumed support is wrong.

Supplementary: Common Errors

Common Error	Correction
Assume mixed strategy means 50–50.	Solve the indifference equations.
Use your own pay-offs to find your probability.	Your mix makes the other player indifferent.
Mix despite a strictly better action.	Only best responses receive positive probability.
Call the opponent “paralysed.”	The opponent is indifferent, not inactive.
Treat changing prices as proof of mixing.	Separate strategic randomisation from demand-based change.
Accept arbitrary payoff numbers.	Ground them in demand, cost, margin, or market position.